

Priyadarshan

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Education

Purdue University, USA <i>PhD in Mechanical Engineering</i> Advisor: Dr. Kevin J. Kircher CGPA: 3.84/4.00	<i>Jan 2022 – Present</i>
Indian Institute of Technology (ISM), India <i>MTech in Mechanical Engineering</i> Advisor: Dr. Subrata K. Ghosh CGPA: 9.50/10.00	<i>Aug 2019 – May 2021</i>
Manipal Institute of Technology, India <i>BTech in Mechanical Engineering</i> Advisors: Dr. Srinivas Shenoy & Dr. Suhas Y. Nayak CGPA: 9.38/10.00	<i>Aug 2013 – May 2017</i>

Experience

Graduate Research Assistant, Purdue University	<i>West Lafayette, IN, USA Jan 2021 – present</i>
National Laboratory of the Rockies	<i>Golden, CO, USA May 2025 – July 2025</i>
Graduate Teaching Assistant, Purdue University: ME 375: Measurements and Controls II ME 597: Distributed Energy Resources ME 308L: Fluid Mechanics Lab	<i>West Lafayette, IN, USA 2022 – 2026</i>
Instructor — ME 200: Thermodynamics, Purdue University	<i>West Lafayette, IN, USA Aug 2025 – Dec 2025</i>

Publications

Journals

7. **Priyadarshan** and K.J. Kircher. Vehicle-to-home discharging halves the impacts of electrification on distribution grids, (*In review*)
6. **Priyadarshan**, C. Crozier, K. Baker and K.J. Kircher. Distribution Grids May Be a Barrier To Residential Electrification, [Cell Reports Sustainability](#), 2025 [🔗](#)
5. E.N. Pergantis, L.D. Premer, A.H. Lee, H. Liu, **Priyadarshan**, E. Groll, D. Ziviani and K.J. Kircher. Protecting residential electrical panels and service through model predictive control: A field study, [Applied Energy](#), 2025 [🔗](#)
4. E.N. Pergantis, **Priyadarshan**, N. Al Theeb, P. Dhillon, J.P. Ore, D. Ziviani, E. Groll and K.J. Kircher. Field demonstration of predictive heating control for an all-electric house in a cold climate, [Applied Energy](#), 2024 [🔗](#)
3. S. Kumar, **Priyadarshan** and S.K. Ghosh. Statistical and Computational Analysis of an Environment-Friendly MWCNT/NiSO₄ Composite Materials, [Journal of Manufacturing Processes](#), 2021 [🔗](#)
2. S. Kumar, **Priyadarshan** and S.K. Ghosh. Statistical and Artificial Neural Network Technique for Prediction of Performance in AlSi10Mg-MWCNT Based Composite Materials, [Materials Chemistry and Physics](#), 2021 [🔗](#)
1. S. Kumar, **Priyadarshan** and S.K. Ghosh. Statistical and Comparative Study of Airborne Particles on New Developed Metal Matrix Composite and Commercial Brake Pad Materials with ANN and Finite Element Analysis, [Computational Particle Mechanics](#), 2021 [🔗](#)

Conferences

6. E.N. Pergantis, L.D. Reyes Premer, A.J. Khabbazi, **Priyadarshan**, F. Wu, D. Ziviani and K.J. Kircher. Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study, [Conference on International Scientific Building and Technology, 2025](#) [🔗](#)
5. E.N. Pergantis, L.D. Premer, A.H. Lee, H. Liu, **Priyadarshan**, E. Groll, D. Ziviani and K.J. Kircher. Protecting residential electrical infrastructure through advanced control: The first field results, [8th International High Performance Buildings Conference at Purdue, 2024](#) [🔗](#)
4. E.N. Pergantis, J. Park, T.J. Bird, **Priyadarshan**, D. Ziviani and K.J. Kircher. Open-source toolbox for identifying envelope dynamics of detached residential buildings, [8th International High Performance Buildings Conference at Purdue, 2024](#) [🔗](#)
3. **Priyadarshan**, E.N. Pergantis, C. Crozier, K. Baker and K.J. Kircher. EDGIE: A simulation test-bed for investigating the impacts of building and vehicle electrification on distribution grids, [Hawai'i International Conference on System Science \(HICSS\), 2024](#) [🔗](#)
2. E.N. Pergantis, L.D. Reyes Premer, **Priyadarshan**, A.H. Lee, P. Dillhon, D. Ziviani, E.A. Groll and K.J. Kircher. Latent and sensible model predictive controller demonstration in a house during cooling operation, [ASHRAE Winter Conference, 2024](#) [🔗](#)
1. E.N. Pergantis, A.S. Sangamnerkar, **Priyadarshan**, J.P. Ore, P. Dillhon, D. Ziviani, E.A. Groll and K.J. Kircher. Sensors, storage and algorithms for practical optimal controls in residential buildings, [ASHRAE Annual Conference, 2023](#) [🔗](#)

In the News

What the effects of electrification could mean for home and car owners, Star City TV 🔗	2025
Purdue offers solutions to blackouts, high energy bills as people electrify their homes and cars, News 🔗	2025
Purdue study: Move to electric homes, vehicles could cost households thousands, WISH-TV 🔗	2025
If every home and personal vehicle went electric in the U.S., power outages could spike unless key measures are taken, Purdue News 🔗	2025
The cost (and savings) of electrifying homes and cars, AP segment aired in Lafayette, Evansville, Monterey-Salinas, and San Francisco, News 🔗	2025
Let go of the thermostat: predictive heating control system demonstrates huge energy savings, Purdue News 🔗	2024

Awards, Fellowships, Achievements

Runner-up in Computational Interdisciplinary Graduate Programs Spring Symposium Doctoral Talk, Purdue University	2026
Bilsland Dissertation Fellow, Purdue University	2026
Perry Graduate Fellowship, Purdue University	2025
1st Position in Jump Into STEM Challenge, National Building Science competition, NLR	2024
William Uffman Fellowship 🔗 , Purdue University	2024
People's Choice Say It in 6 Award 🔗 , Purdue University	2023
Gold Medal - Ranked 1 in Master's Program, IIT Dhanbad	2021
Post-Graduate Research Fellowship, IIT Dhanbad	2019
99th Percentile - Graduate Aptitude Test for Engineers Examination	2019
Ranked Top 5% in Undergraduate program, Manipal Institute of Technology	2017
2nd Position in CAD-A-THON, I.E. Aerospace, Manipal Institute of Technology	2015

Invited Talks

Guest lecturer: Distribution Grids, Mechanical Engineering Purdue University	2026
Guest lecturer: Critical Reading for PhDs, Sustainability Engineering & Environmental Engineering, Purdue University	2025
Lab presentation: EDGIE, Griffin Lab, University of Colorado Boulder	2024

Poster Presentations

Impacts of Residential Electrification, Indiana Sustainability and Resilience Conference, IU Environmental Resilience Institute	2026
Impacts of Home and Vehicle Electrification on Distribution Grids, Grid of Tomorrow Consortium, Purdue University	2025
Characterizing and Mitigating the Impacts of Home and Vehicle Electrification on Distribution Grids, Women in ME Symposium, Purdue University	2023
EDGIE: Emulating Distribution Grids Impact of Electrification, CHPB Conference and IAC Conference, Purdue University	2023

Mentoring

Sophia Evers, BS, Purdue University	2023–2024
Mihir Nair, MS, Purdue University	2023–2024
Khusal, PhD, Purdue University	2023–2024
Srijan Agarwal, BS, IIT Kharagpur	2023

Outreach & Professional Development

Session co-chair, Herrick Conference for Buildings, Purdue University	2026
Vice President, Herrick Student Resource Committee, Purdue University	2024
Sports & Well-being Chair, Official Mechanical Engineering Graduate Association (OMEGA), Purdue University	2023
Social Media Chair, Official Mechanical Engineering Graduate Association (OMEGA), Purdue University	2023
Volunteer, Minority Engineering Program, Purdue University	2023
Placement & Internship Representative, Indian Institute of Technology, Dhanbad	2019–2021
Technical Secretary, I.E. Mechanical, Manipal Institute of Technology	2017
Management Committee, I.E. Mechanical, Manipal Institute of Technology	2016
Event Head, I.E. Mechanical, Manipal Institute of Technology	2016
Volunteer, I.E. Mechanical, Manipal Institute of Technology	2014